PROBLEM CL9

Submarine Propulsion Device: Thrust from the Propeller Power

Original statement · CL9.pdf

Thrust Bernoulli Energy Similarity

OFFICIAL PROBLEM STATEMENT

The submarine propulsion device schematically represented in the figure of CL9.pdf consists of a nozzle, with inlet and outlet cross-section area A_e and A_s, and a propeller (printed as 'fan', corrected by hand to 'propeller') located inside, which imparts a rate of work per unit time (power) $\dot{W}$ to the circulating water stream, whose density and viscosity are ρ and μ, respectively. To determine the thrust T produced when the submarine is at rest and at sufficient depth for gravity to have a negligible effect, please follow these steps:

- By application of the continuity equation determine a relationship between the uniform velocities U_e and U_s at the inlet and outlet sections.
- Use the momentum conservation equation to obtain an expression for the thrust (axial force) that the fluid exerts on the propulsion device as a function of the flow conditions at the inlet and outlet. Assume that the pressure surrounding the device is equal to the ambient pressure p_∞ .
- Assuming that the flow upstream from the entrance is steady and inviscid, use Bernoulli's equation along a given streamline to find a relationship between the flow conditions in the stagnant atmosphere (1) and at the entrance e.
- Use the energy equation to relate the work per unit time $\dot{W}$ with the different fluid variables appearing in the problem. Assume that the variation of internal energy between the inlet and outlet sections is negligible ($e_e = e_s$) and that all of the device walls, including those of the fan, are thermally insulated.
- Combine the above equations to show that the thrust can be expressed in the form $T/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = f(A_e/A_s)$, obtaining the function f.

Source: CL9.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average), incompressible water; gravity negligible at depth (it is absorbed in the reduced pressure $p + \rho g z$, as the official solution notes); uniform properties at e and s.
- Viscous stresses negligible on the sections ($\rho U D/\mu \gg 1$); the exit jet discharges at the ambient pressure, $p_s = p_\infty$; walls and propeller insulated; $e_e = e_s$.
- Data: $[\rho,\ A_e,\ A_s,\ \dot{W},\ p_\infty]$. Unknowns: $[U_e,\ U_s,\ p_e,\ T]$.

Symbol	Meaning	SI unit
A_e, A_s	inlet and outlet areas	m^2
U_e, U_s	inlet and outlet velocities	m/s
p_e, p_∞	inlet and ambient pressures	Pa
$\dot{W}$	power given by the propeller to the water	W
T	thrust	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the submarine (at rest), $\bar{e}_x$ along the axis in the direction of the exit jet. The thrust pushes the device towards $-\bar{e}_x$: $\bar{F} = -T\bar{e}_x$ is the force of the water (inside and outside, at p_∞) on the device.
- Surfaces: Σ_e (inlet, $\bar{n} = -\bar{e}_x$), Σ_s (outlet, $\bar{n} = +\bar{e}_x$), Σ_w (nozzle walls, fixed) and Σ_p (propeller, moving).

$$\boxed{\bar{F} = -T\,\bar{e}_x = \int_{\Sigma_w,\Sigma_p} \left[(p - p_\infty)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

(1) Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \, (\bar{v} - \bar{v}_c) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \, (\bar{v} - \bar{v}_c) \cdot \bar{n} \, d\sigma$
Σ_e	$-\rho U_e A_e$
Σ_s	$+\rho U_s A_s$
$\Sigma_w,\ \Sigma_p$	0

$$\boxed{U_e A_e = U_s A_s} \qquad (1)$$

(2) Linear momentum (Notes Eq. 3.35)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \bar{v} \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \bar{v} \, [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} (p - p_\infty) \, \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma + \underbrace{\cancel{\int_{V_c} \rho \bar{g} \, dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \bar{v} \, [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma$	$-\int (p - p_\infty) \, \bar{n} \, d\sigma$	$\int \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma$
Σ_e	$-\rho U_e^2 A_e \, \bar{e}_x$	$(p_e - p_\infty) A_e \, \bar{e}_x$	0
Σ_s	$+\rho U_s^2 A_s \, \bar{e}_x$	$0 \ (p_s = p_\infty)$	0
$\Sigma_w,\ \Sigma_p$	0	$-\int (p - p_\infty) \, \bar{n} \, d\sigma$	$\int \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma$

Boxed terms: $-\bar{F} = T\bar{e}_x$. With (1), $\rho U_s^2 A_s - \rho U_e^2 A_e = \rho U_e A_e (U_s - U_e)$.

$$\rho U_e A_e (U_s - U_e) = (p_e - p_\infty) A_e + T \implies \boxed{T = \rho U_e A_e \,(U_s - U_e) - (p_e - p_\infty) A_e} \quad (2)$$

(3) Bernoulli upstream of the inlet

Steady inviscid flow from the stagnant water far ahead (1), where $v = 0$ and $p = p_\infty$, to the inlet e (Notes Eq. 4.17 with the reduced pressure):

$$p_\infty = p_e + \tfrac{1}{2}\rho U_e^2 \implies \boxed{p_e - p_\infty = -\tfrac{1}{2}\rho U_e^2} \quad (3)$$

`(4) Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \bar{v}_c)\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\, d\sigma + \underbrace{\int_{V_c} Q_R\, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	$-\int \bar{q}\cdot\bar{n}\, d\sigma$
Σ_e	$-\rho U_e A_e\left(e_e + \frac{U_e^2}{2}\right)$	$+p_e U_e A_e$	0	0
Σ_s	$+\rho U_s A_s\left(e_s + \frac{U_s^2}{2}\right)$	$-p_\infty U_s A_s$	0	0
Σ_p	0	$-\int_{\Sigma_p} p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int_{\Sigma_p} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	0 **(insulated)**
Σ_w	0	0	0	0 (insulated)

Boxed Σ_p terms: the propeller power $\dot{W}$. With $e_e = e_s$, $\rho U_e A_e = \rho U_s A_s = \dot{m}$ and (3):

$$\dot{W} = \dot{m}\left[\frac{p_\infty - p_e}{\rho} + \frac{U_s^2 - U_e^2}{2}\right] = \dot{m}\left[\frac{U_e^2}{2} + \frac{U_s^2 - U_e^2}{2}\right] \implies \boxed{\dot{W} = \tfrac{1}{2}\rho A_s U_s^3, \qquad U_s = \left(\frac{2\dot{W}}{\rho A_s}\right)^{1/3}} \quad (4)$$

(5) Thrust in similarity form

Substituting (3) into (2) and using (1), $U_e A_e = U_s A_s$ and $U_e = U_s A_s / A_e$:

$$T = \rho U_e A_e (U_s - U_e) + \tfrac{1}{2}\rho U_e^2 A_e = \rho U_e A_e \left(U_s - \frac{U_e}{2}\right) = \rho A_s U_s \left(U_s - \frac{U_s}{2}\frac{A_s}{A_e}\right) = \rho A_s U_s^2 \left(1 - \frac{1}{2}\frac{A_s}{A_e}\right).$$

With (4), $\rho A_s U_s^2 = \rho A_s \left(\frac{2\dot{W}}{\rho A_s}\right)^{2/3} = 2^{2/3}\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}$, so

$$\boxed{\frac{T}{\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}} = f\left(\frac{A_e}{A_s}\right) = 2^{2/3}\left(1 - \frac{1}{2\,A_e/A_s}\right)} \qquad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- f grows with A_e/A_s: a large intake (slow inflow, small suction depression at the inlet) gives the most thrust for a given power, tending to $f \to 2^{2/3} \approx 1.587$ as $A_e/A_s \to \infty$. For $A_e = A_s$, $f = 2^{2/3}/2 \approx 0.794$.
- Thrust grows only as $\dot{W}^{2/3}$: doubling the power gives $2^{2/3} \approx 1.59$ times the thrust, because the extra power goes into a faster jet whose kinetic energy is wasted.
- Dimensions (SI): $[\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}] = (\mathrm{kg/m^3})^{1/3}(\mathrm{kg\,m^2/s^3})^{2/3}(\mathrm{m^2})^{1/3} = \mathrm{kg\cdot m/s^2} = \mathrm{N}$, so f is dimensionless.
- Numerical check (hypothetical data): $\rho = 1000$ kg/m^3, $A_s = 0.05$ m^2, $A_e = 0.1$ m^2, $\dot{W} = 50$ kW give $U_s = 12.599$ m/s, $U_e = 6.2996$ m/s, $T = 5952.8$ N, and $T/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = 1.1906 = f(2)$; the energy balance (4) gives back $\dot{W} = 50$ kW.
- Agreement with the official solution: CL9.pdf obtains (1) to (5), $f = 2^{2/3}\left(1 - \tfrac{1}{2}A_s/A_e\right)$.